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Comparison, advancement, and performance evaluation of heat exchanger assembly in solid-state hydrogen storage device

R. Sreeraj, A.K. Aadhithiyan, S. Anbarasu *

Hydrogen Energy Systems Research Facility, Department of Mechanical Engineering, NIT Rourkela, Odisha, 769008, India

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ABSTRACT

Despite myriads of mathematical and experimental studies using various heat exchanger-metal hydride assemblies, an adept comparison of reactors on standard performance defining parameters, i.e., reactor weight, external energy utilization, and alloy material, are not studied or reported for industrial storage containers and onboard applications. This lacuna is a critical barrier in developing efficacious hydrogen storage reactors for various applications. This article chronicles the 3-D design and optimization of three basic heat exchanger setups of solid-state hydrogen storage systems (shell and tube, spiral tube, and tubular) on critical parameters. For 5 kg $LaNi_5$, a tubular design with small tube diameters has resulted in a shorter reaction time (120 s) for 90% saturation at 303 K and 15 bar inlet conditions; however, the reactor's gravimetric density was exceptionally high. In contrast, the reactor with three spiral tubes accomplished an exemplary reaction rate at a reasonable gravimetric density. This spiral arrangement became the base for developing a novel reactor that integrated heat pipes to reduce the sorption time. The proposed design achieved a specific output energy rate of 527 W/kg for 90% saturation in 366 s at a supply pressure of 15 bar, which is 23.7% higher than the reactor without heat pipes.

1. Introduction

Secure, efficient, and cost-effective hydrogen storage is mandated for marketing hydrogen application systems. Metal hydride-based hydrogen storage (MHHS) techniques could be a viable alternative to the high-pressure or high-cost hydrogen storage methods currently in use [1]. Nevertheless, controlling the reaction heat due to the hydrogenation or dehydrogenation process (exothermic absorption and endothermic desorption) is ineluctable [2–4]. Considerable efforts have been made in recent years to increase the reaction rate by effectively improving the heat transfer from/to metal hydride by enhancing the thermophysical properties of the hydride [5–9] or by increasing the heat transfer surface area [10–14], or heat transfer coefficient of heat transfer fluid (HTF) [15–21].

A prominent heat exchanger design is critical for MHHS devices to maximize alloy content while minimizing empty reactor weight and volumetric density [22]. Shell and tube, spiral, and tubular heat exchangers are the most conventional heat exchanger designs used in MHHS systems [3]. Few authors have evaluated the effects of shell and tube heat exchanger assembly in metal hydride bed (MHB), wherein heat transfer fluid circulates through tubes within a metal hydride-filled casing. Anbarasu et al. investigated the absorption and desorption rates of 2.75 kg $LmNi_{4.91}Sn_{0.15}$ using 36 and 60 straight embedded tubes, respectively. The desorption time was 1320 s for 36 tubes at a fluid inlet temperature of 60 °C and the absorption time was 720 s at 30 °C and 25 bar inlet conditions. Simultaneously, the reactor with 60 tubes took 660 and 540 s for the same operating conditions [23,24]. Kumar et al. achieved an absorption time of 300 s for 40 kg $LaNi_{4.7}Al_{0.3}$ using 99 straight tubes at 40 bar supply pressure [25]. Similarly, in the tubular bundle configuration, HTF flows through the shell set in with alloy-filled tubes. At a peak supply pressure of 13.8 MPa and an inlet heat transfer oil temperature of 130 °C, a reactor containing 86 kg of modified sodium alanate in a stacked array of 48 tubes stored 3 kg of hydrogen in 30 min [26]. In the spiral tube design, HTF circulates through spiral tubes embedded within the alloy-filled casing. 1 kg of $LaNi_5$ with a single helical tube integrated with copper fins completed hydrogenation in 500 s at a supply pressure and cooling fluid inlet temperature of 10 bar and 20 °C, respectively [27]. Visaria and Mudawar, on the other hand, accomplished the desorption of 4 kg high-pressure metal hydride ($Ti_{1.1}CrMn$) equipped with a spiral heat exchanger assembly in 1440 s at a temperature of 2.5 °C [28]. Here the time taken for the absorption process at a supply pressure of 270 bar was 1260 s and 720 s at an HTF

* Corresponding author.
E-mail address: anbarasus@nitrkl.ac.in (S. Anbarasu).

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temperature of 21 and 2.5 °C, respectively.

Raju and Kumar [3] optimized the shell and tube, helical, and tubular bundle reactors based on geometrical parameters, concluding that the helical coil heat exchanger in MHB provides better gravimetric and volumetric densities (0.0139 kg/kg and 0.0167 kg/L). However, the study falls short of thoroughly examining the thermal and mass transfer aspects, as well as a proper comparison. Mellouli et al. [29] compared a tubular reactor with an outer cooling jacket to a spiral tube design with and without an outer cooling jacket. The spiral design with an outer cooling jacket completed 80% absorption in 750 s for 1 kg $LaNi_5$ at 10 bar hydrogen pressure and 293 K inlet HTF temperature, which is 70% faster than the tubular reactor. Liang et al. investigated the performance difference between a metal hydride reactor with heat transfer frames such as a straight tube, spiral, its combo, and dual spiral tube and found that the dual spiral heat exchanger metal hydride assembly absorbed 5.19 g of hydrogen in 271 s, whereas the straight tube takes 1321 s for 5.52 g.

Although multifarious studies have reported on MHHS, it is evident from the literature review that there has been no unified and comprehensive comparison of these storage systems. Furthermore, the existing research area lacks in-depth studies of configurations with similar hydride materials, gravimetric capabilities, and operating conditions, limiting the selection of viable models for diverse applications [30,31]. The present work is focused on developing a metal hydride heat exchanger assembly that can effectively control thermal variation in hydride beds to reduce the reaction time while offering a higher specific output energy rate at a less gravimetric penalty. The MHB combined with the described pre-set heat exchanger designs is simulated for different geometric parameters to evaluate its absorption performance. Further, the appropriate bed layout is chosen based on the lesser reactor weight, higher specific output energy rate and quick absorption time that satisfies the optimum performance criteria for a 5 kg metal hydride reactor. The optimal design is further improvised by enhancing the heat transfer in the bed by mounting heat pipes, and then its hydrogen sorption performance at various operating conditions is assessed, which enumerates the novelty of the design and work.

2. Mathematical model

All 3D models in this analysis used 5 kg of $LaNi_5$–H_2 and were solved with COMSOL Multi-physics® 5.6. Heat transfer through porous media, heat transfer in solids, turbulent flow, laminar flow, heat transfer in fluids, and darcy law were the physics interfaces used for numerical simulations, which were solved using linear solvers such as PARDISO for the temperature of porous, solid, and fluid domains and Algebraic Multigrid GMRES for fluid velocity. Certain assumptions were crucial to expedite the current mathematical procedure due to the complexities of reaction kinetics. The solution methodology and assumptions used to simplify the governing equations are similar to those reported by Anbarasu et al. [32]. Furthermore, the influence of the reactor wall on the storage system during hydrogenation was neglected in the numerical procedure. Based on assumptions made [32], the energy conservation equation for the suggested porous medium is as follows:

$$(\rho C_P)_{eff}\frac{\partial T}{\partial t}+(\rho C_P)_g\left(\overrightarrow{u_g}\cdot\nabla T\right)=\nabla\left(k_{eff}\cdot\nabla T\right)+S\cdot\frac{\nabla H}{M_g} \quad (1)$$

The volume-averaged method for specific heat capacity and thermal conductivity of porous alloy is

$$\begin{aligned}(\rho C_P)_{eff} &= \varepsilon\cdot(\rho C_P)_g+(1-\varepsilon)(\rho C_P)_{MH}\ and\\ k_{eff} &= \varepsilon\cdot k_g+(1-\varepsilon)k_{MH}\end{aligned} \quad (2)$$

where S is the hydrogen absorption rate per unit volume, and it can be expressed as follows

$$S=C_a\exp\left(-\frac{E_a}{R_u.T}\right)ln\left(\frac{P}{P_{eq}}\right)(\rho_{sat}-\rho_{MH})(1-\varepsilon) \quad (3)$$

where the equilibrium pressure can be calculated from the Van't Hoff equation and it can be expressed as,

$$P_{eq}=P_{ref}\exp\left(-\frac{\Delta H}{R_u.T}+\frac{\Delta S}{R_u}+(\varphi+\varphi_0)\tan\left(\pi\left(\frac{wt\%_t}{wt\%_{sat}}-0.5\right)\right)+\frac{\beta}{2}\right) \quad (4)$$

The velocity and permeability of hydrogen gas through the porous hydride can be written by Darcy's law and Kozeny-Carman equation [32,33], where permeability is estimated to be 1.08×10^{-12} m^2

$$\overrightarrow{u}_g=-\frac{\kappa}{\mu_g}\cdot\nabla P\ and\ \kappa=\frac{1}{150}\frac{d_p^2\varepsilon^3}{(1-\varepsilon)^2} \quad (5)$$

Whereas Eq. (6) gives the mass conservation of metal hydride

$$(1-\varepsilon)\frac{\partial\rho_{MH}}{\partial t}=S \quad (6)$$

During the absorption process, water as HTF is supplied to maintain the reaction in thermal equilibrium, and the energy equation for HTF is

$$(\rho C_P)_f\frac{\partial T}{\partial t}+(\rho C_P\overrightarrow{u})_f\cdot\nabla T=\nabla\left(k_f\cdot\nabla T\right) \quad (7)$$

At the beginning of the sorption process, temperature, pressure and density all over the MHB are assumed to be uniform throughout the alloy bed.

$$\left.T=T_i=T_{in,f};\ P=P_i=P_s;\ \rho=\rho_i;\ wt\%=0\right\}at\quad t=0 \quad (8)$$

The initial temperature of MHB can be considered the inlet temperature of HTF, and the pressure of the metal hydride reaches the supply pressure almost immediately; hence, the initial pressure can be considered the supply pressure of hydrogen gas.

At the cooling tube and HTF interface:

$$q=k\frac{\partial T}{\partial r}=h_f\left(T_w-T_f\right)\ t\geq0 \quad (9)$$

At the Porous medium and filter wall interface:

$$q=\frac{\partial T}{\partial r}=0;P=P_s\ t\geq0 \quad (10)$$

3. Model description and design of different MHHS devices

In this present study, three prominent heat exchanger designs of metal hydride-based energy storage studies were explored to propose a simple, compact, and efficient energy storage device. The reaction kinetics of AB_5 metal hydride was investigated using reactors comprising embedded straight tubes (shell-and-tube design), spiral tubes, and tubular frameworks (Fig. 1).

Although numerous factors and arrangements exist for optimizing envisaged metal hydride reactors, only a few core indices are used to evaluate the proposed investigation. The thickness of casings and tubes was kept uniform throughout the study (except for tubular design), and the selection was based on mechanical factors such as bearing pressure and material properties. Maximum pressure of 100 bar and a safety factor of 3 were accounted for in calculating tube and shell thickness. Stainless steel material of grade SS316L was used for all structures due to cost implications and exceptional thermal properties. Other parameters, including HTF's inlet temperature and flow rate, were maintained constant for the optimization study; temperature and flow rate was set to 303 K and 0.5 l/min, respectively. Depending on the Reynolds number, the flow of HTF through the heat exchanger tubes is categorized as laminar or turbulent. The HTF flow is determined by the k-ε model for fully turbulent conditions. Moreover, on the exterior of the reactor wall, a heat transfer coefficient of 500 W/m^2K was given. The thermophysical

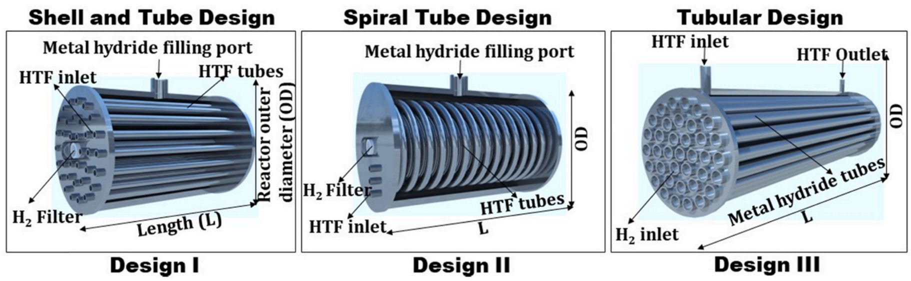


Fig. 1. Visualization of the heat exchanger assemblies used, (i) shell and tube, (ii) spiral tube, and (iii) tubular design.

properties of materials and other parametric values used in the numerical model are summarized in Table 1.

3.1. Design I – shell and tube design

In shell and tube heat exchanger assembly, the hydride is filled inside a cylindrical casing, and HTF is circulated through embedded straight tubes within the cylinder to regulate hydride bed temperature. The arrangement of cooling tubes is kept with an equal separation between them for better performance [34]. The cooling tubes inside the shell are arranged in three circular strata of varying radial distances, and the number of tubes used is 24, 36, 48, and 60. The outer diameter and thickness of the shell are 114.3 mm and 5.85 mm, respectively, and the cooling tube diameter is varied from 6.35 to 12.7 mm for analysis. Fig. 2 (a) depicts a cross-sectional view of the design, and Table 2 reveals variation in physical parameters as well as the computed weight ratio of alloy (mass of alloy to total mass of reactor). Due to symmetrical manifestation, a 3D simulation of the reactor's semi portion is considered for analyzing the heat and hydrogen transfer. A Nusselt number correlation is shown in Eqs. 11–13; this helps to calculate the heat transfer coefficient of HTF flowing through tubes [34].

$$Nu = 1.86\left(\mathrm{Re}\cdot\mathrm{Pr}\frac{d_{CT}}{L_{CT}}\right)^{1/3} \quad \textit{Laminar flow}\ (Re < 2100) \tag{11}$$

$$Nu = 0.116\left[1+\left(\frac{d_{CT}}{L_{CT}}\right)^{\frac{2}{3}}\right]\left(\mathrm{Re}^{\frac{2}{3}}-125\right)(\mathrm{Pr})^{1/3} \quad \textit{Transition regime}\ (2100 < Re < 10,000) \tag{12}$$

$$Nu = 0.023\left[1+\left(\frac{d_{CT}}{L_{CT}}\right)^{0.7}\right](\mathrm{Re})^{0.8}(\mathrm{Pr})^{1/3} \quad \textit{Turbulent flow}\ (Re > 10,000) \tag{13}$$

Table 1
The thermophysical properties of $LaNi_5$ and hydrogen, as well as other parameters adopted in the numerical model [12].

Property	Values		
	$LaNi_5$	H_2	Steel
Crystal Density, ρ (kg/m^3)	8200.0		7850.0
Porosity, ε	0.5		
Thermal conductivity, k (W/m K)	2.4	0.18	19.0
Specific heat capacity, C_p (J/kg K)	419.0	14890.0	490.0
Molecular weight, M (kg/mol)	0.432	0.002432	
Maximum H_2 capacity, wt%	0.014		
Slope factor, φ	0.038		
Slope constant, $φ_o$	0.0		
Hysteresis factor, β	0.137		
Activation energy, E_a (J/mol H_2)	21179.6		
Reaction entropy, ΔS (J/mol H_2 K)	108.0		
Reaction enthalpy, ΔH (J/mol H_2)	30800.0		
Reaction constant, C_a (1/s)	59.187		

3.2. Design II – spiral design

In design II, the spiral tubes enhance the heat transfer rate within the metal hydride-filled cylindrical shell. Spiral heat exchangers outcompete other heat exchangers in terms of heat transfer coefficient. The general graphical representation of the design is depicted in Fig. 2 (b). The number of spiral tubes, axial pitch, diameter, and radial distance are varied for design optimization. Table 3 elucidates the variation in physical parameters of the design at each arrangement. The remaining dimensions and parameters are identical to design I. The prevalent equation of the Nusselt number for calculating the heat transfer coefficient of a spiral tube heat exchanger is given in Eq. (14) [35].

$$Nu_h = \frac{h_f d_{CT}}{k_f} = 0.0266\left(\frac{\mathrm{Re}^{0.85}}{i^{0.15}} + 0.225 i^{1.55}\right)\mathrm{Pr}^{0.4} \tag{14}$$

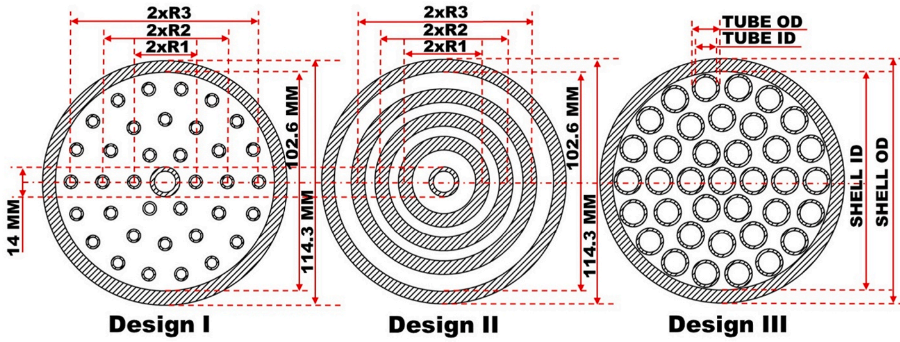
3.3. Design III – tubular design

In this heat exchanger design, metal hydride-filled tubes are housed within the cylindrical shell (Fig. 2 (c)). Apart from tube diameter, no other parameters are optimized in this arrangement, consisting of 37 tubes in each case. Since no internal heat transfer methods are used in metal hydride tubes, the tube diameter was limited to 25.4 mm. The time required to reach hydrogen saturation for tubes larger than 25.4 mm in diameter was exceptionally long. The physical parameters of each setup are tabulated in Table 4. The Nusselt number correlation of HTF flowing over the banks of tubes is achieved using the same equation used for the shell and tube design (Eqs. 11–13).

4. Grid independent test and model validation

The numerical analysis's 3D model was tested for various mesh sizes to optimize the effect of element sizes on the average hydrogen concentration and bed temperature during the absorption process. For all domains in the reactor, the grid element size varied from coarse to extra fine (Table 5). It was observed that the fluctuation in numerical results of fine and extra fine mesh was almost statistically insignificant when contemplating the computational time. As a result, the fine mesh was used for every simulation throughout the study.

As shown in Fig. 3, the mathematical model used in this study was validated with an experimental result of a shell and tube reactor design from Karmakar et al. [36]. In the experiment, the author used 10 kg of $LaNi_5$ at a supply pressure of 25 bar and HTF temperature of 298 K with a fluid flow rate of 20 l/min. It took 1700 s to reach 100% hydrogen saturation for these operating conditions. Fig. 3 demonstrates a good match between the numerically predicted hydrogen concentration and the experimental results. The underlying assumptions for the thermophysical properties of the alloy are primarily responsible for the subtle differences in results.



(a) (b) (c)

Fig. 2. Graphical representation of design configurations used for the simulation.

Table 2
Design specifications of different cases of metal hydride-shell and tube design.

cases	Radial distance (mm)									
	R1	R2	R3	Tube OD (mm)	Tubes (nos.)	L (mm)	Tube mass (kg)	Total mass (kg)	Alloy weight ratio	SCR(stL/min.g)
case 1	14.36	29.15	43.92	6.35	24	165.15	0.474	8.12	0.615	0.014
case 2	14.36	29.15	43.92	6.35	36	174.22	0.751	8.55	0.584	0.025
case 3	14.36	29.15	43.92	6.35	48	184.34	1.059	9.02	0.554	0.033
case 4	14.36	29.15	43.92	6.35	60	195.72	1.406	9.55	0.523	0.043
case 5	18.08	29.15	40.23	6.35	36	174.22	0.751	8.55	0.584	0.028
case 6	11.43	29.15	46.87	6.35	36	174.22	0.751	8.55	0.584	0.019
case 7	11.41	20.23	33.47	6.35	36	174.22	0.751	8.55	0.584	0.021
case 8	24.65	37.89	46.71	6.35	36	174.22	0.751	8.55	0.584	0.010
case 9	14.36	29.15	43.92	9.53	36	219.52	1.497	10.02	0.498	0.041
case 10	14.36	29.15	43.92	12.7	36	344.53	4.360	14.89	0.335	0.076

Table 3
Design specifications of different cases of metal hydride-spiral tube design.

cases	Radial distance (mm)											
	R1	R2	R3	R4	Radial Pitch (mm)	Tube OD (mm)	Tube (nos.)	L (mm)	Tube mass (kg)	Total mass (kg)	Alloy weight ratio	SCR(stL/min.g)
case 1	21.77	36.53			10	6.35	2	166.28	0.509	8.18	0.611	0.024
case 2	14.36	29.15	43.92		10	6.35	3	190.82	1.257	9.32	0.536	0.040
case 3	15.86	24.72	33.58	42.44	10	6.35	4	210.14	1.846	10.22	0.489	0.051
case 4	14.36	29.15	43.92		8	6.35	3	204.90	1.686	9.97	0.501	0.048
case 5	14.36	29.15	43.92		12	6.35	3	182.46	1.002	8.93	0.559	0.035
case 6	14.36	29.15	43.92		15	6.35	3	174.82	0.769	8.57	0.582	0.025
case 7	14.36	29.15	43.92		12	9.53	3	251.82	2.189	11.23	0.445	0.063
case 8	14.36	29.15	43.92		15	9.53	3	221.69	1.544	10.10	0.494	0.057
case 9	14.36	29.15	43.92		15	12.7	3	354.21	3.373	14.06	0.355	0.078
case 10	18.08	29.15	40.23		12	6.35	3	182.47	1.002	8.93	0.559	0.036
case 11	11.43	29.15	46.87		12	6.35	3	182.49	1.003	8.93	0.559	0.026
case 12	11.41	20.23	33.47		12	6.35	3	178.88	0.850	8.76	0.570	0.020
case 13	24.65	37.89	46.71		12	6.35	3	194.98	1.337	9.51	0.525	0.024

Table 4
Design specifications of different cases of metal hydride-tubular design.

cases	Tube OD (mm)	thickness (mm)	No. Tubes	L (mm)	Tube mass (kg)	Total mass (kg)	Shell OD (mm)	Shell ID (mm)	Alloy weight ratio	SCR(stL/min.g)
case 1	12.7	1.245	37	397.92	5.175	16.56	114.3	102.26	0.301	0.070
case 2	15.88	1.651	37	262.19	5.617	16.32	141.3	128.19	0.306	0.060
case 3	19.05	1.245	37	151.26	3.058	12.33	168.28	154.05	0.405	0.044
case 4	25.4	2.1082	37	92.43	4.139	13.07	219.08	202.72	0.382	0.037

Table 5
The detailed different mesh sizes used for grid-independent test.

Sl. No.	Grid name	Number of elements	Maximum element size	Minimum element size
1	Coarse	795161	5.72	1.71
2	Normal	1036146	3.83	1.14
3	Fine	1291115	3.03	0.572
4	Extra fine	7268580	1.31	0.0857

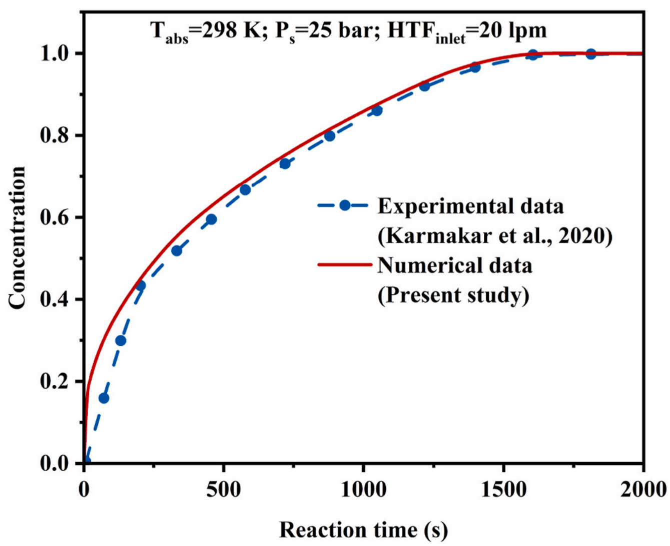


Fig. 3. Validation of the mathematical model with experimental data.

5. Results and discussion

5.1. Comprehensive analysis and optimization of metal hydride reactor designs

The design configurations of MHB with three potential heat exchanger layouts used for the comparative analysis are depicted in Figs. 1 and 2. The geometrical parameters of these designs are interactively varied for determining the optimized arrangement from each. The optimal design choice is based on including and organizing the heat transfer frames to augment the heat transfer rate most evenly from the considered MHB structure. Table 2 summarizes the various shell-and-tube design cases based on physical criteria, alloy weight ratio, and specific charging rate (SCR), which is defined as the ratio of H_2 absorption rate for 90% saturation per unit mass of alloy ($\frac{q_H}{t_a W_a}$, stL/min.g). Similarly, data for spiral tube and tubular designs are provided in Tables 3 and 4, respectively. The 3D designs of these frameworks are thoroughly examined based on the reaction time and the drop of average bed temperature for 90% saturation, as illustrated in Fig. 4 (a)–(c). The figure conjectures that saturation time and temperature reduction are inversely proportional to the gravimetric and volumetric density of the reactor. Reducing the saturation time by increasing the heat transfer area will lead to higher volumetric and gravimetric penalties. Hence, selecting an optimum configuration based on any of these single criteria is not an ideal option.

It is inevitable to choose an optimized arrangement using a comprehensive index that incorporates the heat recovered, weight, and time constraints. Specific output energy rate (SOER, W/kg) is defined as the ratio of total heat recovery rate for 90% hydrogen saturation to the total weight of the alloy. It specifies the efficacy of the structured heat exchanger assembly in the metal hydride bed, i.e., the maximum energy per unit time that can be extracted by the HTF per unit hydride mass. This term describes the effectiveness of the reactor for a specific weight ratio of alloy. A reactor with a high SOER and alloy weight ratio is the archetype for an optimized design, and the equation for SOER is:

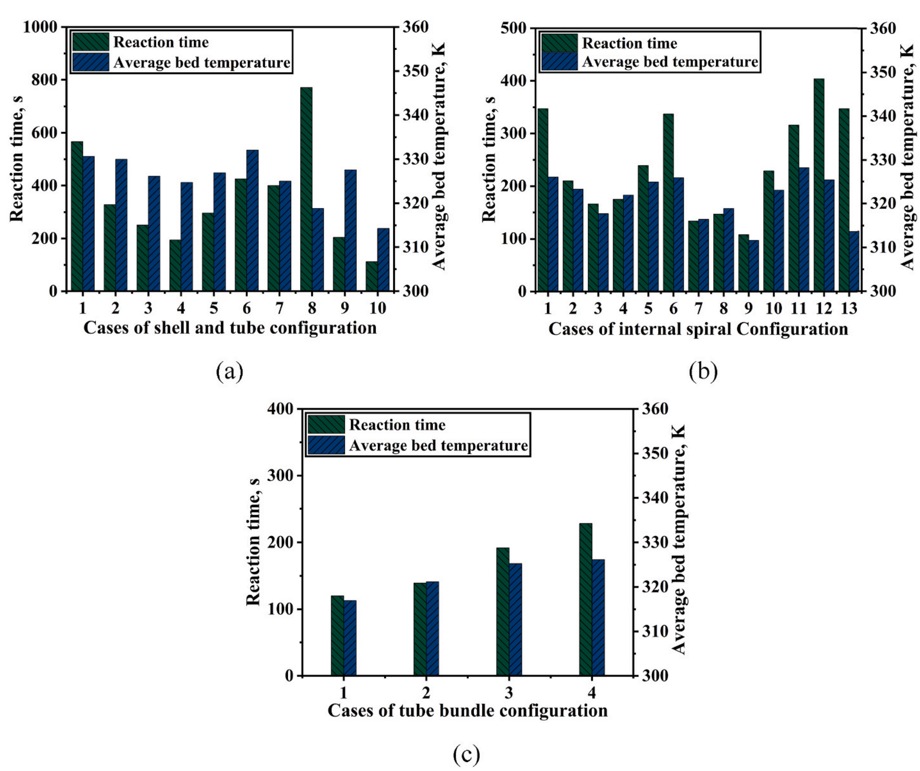


Fig. 4. (a)–(c) Reaction time and average bed temperature at 90% saturation for various adopted MHB configurations.

$$SOER = \frac{E_{out}}{t_a W_a} \tag{15}$$

where E_{out} and t_a are the total heat extracted from the reactor by the HTFs, and the time taken for 90% saturation of the MHB, respectively and W_a represents the mass of the alloy.

The quantity E_{out} can be evaluated by the following equation.

$$E_{out} = \int_{t=0}^{t_a} (\rho u A C_p)_f \left(T_{f,outlet} - T_{f,inlet}\right) + h_{air} A_c (T_c - T_{air}) + h_{hp} A_{hp} \left(T_a - T_{hp}\right) \tag{16}$$

Here the first term represents heat extracted by the heat exchanger, the second term convective heat transfer from the reactor's external periphery, and the final term is only valid where heat transfer via heat pipes is possible. The temperature for each domain is calculated by taking the surface average of the respective boundaries.

The performance evaluation of all cases of each design is shown in Figs. 4 and 5. It consists of the factors such as reaction time and average bed temperature at 90% hydrogen saturation and their respective SOER and alloy weight ratio. These are the main parameters to consider when selecting an appropriate design. For industrial-scale stationary storage applications, the weight ratio is not as important as other factors. Here the reaction time, average bed temperature, and SOER are related to the reaction kinetics of metal hydride, and the weight ratio of the alloy is based on the physical consideration of the reactor. For the design I, case 3 can be presumably considered as the optimized design from the arrangements shown in Table 2, as the performance of other designs is not optimally favoured (Fig. 4 (a)); here, 48 HTF tubes having an outer diameter of 6.35 mm were used, the tube stack layer nearer to the filter has 6 tubes, and the middle layer has 18 tubes, and the last layer consists of 24 tubes. This design has comparatively less saturation time with an alloy weight ratio higher than 0.5. Case 3 (Table 2) has achieved a 90% saturation time at 251 s with a specific charging rate of 0.0337 stL/min.g, and the temperature drop at the instance is 326 K. As the time and temperature for the 90% saturation level are unsatisfactory for design selection, the combined performance indices evaluation of SOER and alloy weight ratio of all cases (Design I) are specified in Fig. 5 (a). In the manuscript, the minimum alloy weight ratio required for design selection is set at 0.5. As shown in Fig. 5 (a), case 3 (Table 2) is superior in that it has a SOER of 740.1 W/kg at an alloy weight ratio of 0.554 for design I. The SOER of each case mentioned in Table 2 is 322.7, 559.9, 740.1, 968.8, 624.4, 430.4, 460.4, 243.6, 924.6 and 1795.7 W/kg, respectively.

Similarly, 13 cases were studied for design II, shown in Table 3. Due to the higher heat transfer coefficient, this design performs better than the initial arrangement (Design I) with comparatively less system weight. The time for reaching 90% saturation and respective temperature for all cases were shown in Fig. 4 (b). The optimized design can be put forward as case 10 (Table 3); here, the radial distance between the spiral tubes is 18.08, 29.15, and 40.23 mm, respectively. The alloy weight ratio of this reactor is 0.559, and the length of the reactor to fill 5 kg alloy is 182.47 mm. Here, the alloy weight ratio of the reactor is nearly equivalent to that of a shell and tube design with 48 HTF tubes, but it has a 9.5% higher SCR of 0.0369 stL/min.g. For the chosen design, the time taken for 90% hydrogenation is 229 s at a temperature of 323 K. Even though there is a reduction of 8.7% in saturation time compared to the shell and tube design, the difference in temperature drop is indiscernible; which is due to the comparatively less temperature variation of $LaNi_5$ with concentration. The SOER for each case in design II is 523.5, 862, 1802.5, 1138.2, 789.5, 558, 1615.7, 1406.9, 2085.9, 828.9, 591.9, 476, and 565.6 W/kg. Considering the alloy weight ratio, case 10 (Table 3) with a SOER of 828.9 W/kg is the prerogative choice; in other cases, either the alloy weight ratio is less than 0.55, or the SOER is inadequate. From the collective analysis of designs I and II, the spiral design (case 10) improves the heat recovery rate by 12% compared to the shell and tube design (case 3).

The third design has been optimized based on the diameter of the tube in which the alloy is filled. The minimum tube diameter has been set at 12.7 mm because the further reduction of the tube diameter

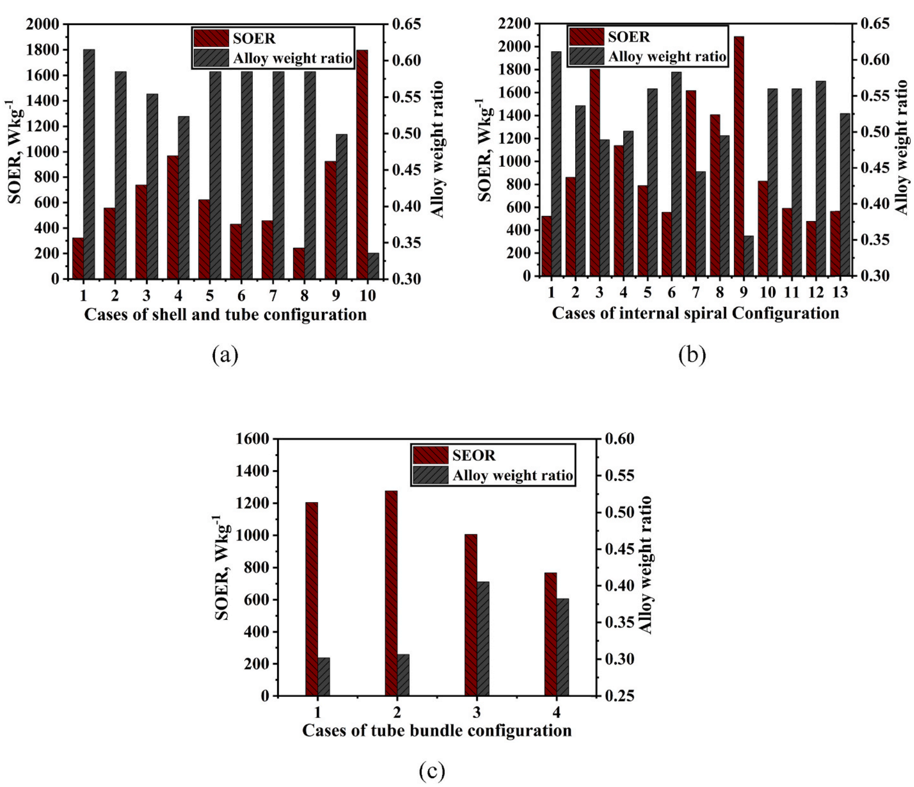


Fig. 5. (a)–(c) SOER and alloy weight ratio for different MHB configurations.

dissuades the practicality of MHHS devices. Here, 4 cases were studied, which are summarized in Table 4. The time and temperature for 90% saturation of each case are shown in Fig. 4 (c). The heat transfer enhancement of such a design is reasonably good, but the weight ratio of the reactor is meagre, especially at a lower diameter, making it unviable for energy storage applications. Despite having higher SOER values such as 1204.8, 1276.8, 1006.2, and 766.8 W/kg for design III, the alloy weight ratio in all cases is below 0.5 (Fig. 5 (c)).

5.2. Design progression

Based on an evaluation of the various heat exchanger assemblies, it was discovered that the spiral tube design with three spiral tubes at radial distances of 18.08, 29.15, and 40.23 mm outperformed other designs when both SOER and specific alloy weight percentages were factored together. Hence, this design was selected and upgraded by appending heat pipes between the outermost tube and the reactor's external surface, as shown in Fig. 6. Table 6 outlines the characteristics of the heat pipe used in the present analysis. Adding heat pipes between the reactor's outer surface and the outermost spiral tube will increase the reaction rate since the hydrogenation volume is greater in the vicinity of the external spiral tube than in the internal tubes. In in-progress studies (parametric analysis), an adiabatic boundary condition is carried out to determine the actual effect of heat pipes on reaction kinetics.

5.2.1. Effect of number of heat pipes

The effect of coupling heat pipes to a spiral design on heat and hydrogen mass transfer is depicted in Fig. 7. The HTF inlet temperature was set to 303 K with a flow rate of 0.5 l/min and the hydrogen supply pressure through the filter was considered 15 bar. For each study, the maximum hydrogen storage capacity value was about 1.4 wt%. Except for the parameters used for the specified analysis, all other conditions are presumed the same for the subsequent studies. The position of heat pipes was not accounted for in design optimization; however, the most susceptible part of the reactor is nearer to the outermost spiral tube, where the pressure drop is high and the heat transfer rate is relatively low compared to other tubes; thus, between the last spiral tube and the reactor's outer periphery is found to be the appropriate location for heat pipes. The absorption time for the reactor without considering heat pipes are 445 s and for the reactor with two heat pipe is 406 s, i.e., there was a gradual reduction in the reaction time on increasing the number of heat pipes, the reactor with four and six heat pipe the absorption time is reduced to 366 and 330 s respectively. The reduction in reaction time was due to the addition of heat pipes as a heat transfer simulator. The number of heat pipes in the current analysis was limited to six due to cost considerations in practical applications. The SOER of the reactor without heat pipes are 426 W/kg, while the SOER is improved to 473 W/kg for two heat pipes — an increase of 11%. The same dynamics are maintained as the number of heat pipes increases; a reactor with four heat pipes yielded a SOER of 527 W/kg, while the reactor with six heat pipes has 585 W/kg. The benefits of adding heat pipes to a reactor with heat transfer from the outside surface through other means are statistically insignificant. The number of heat pipes was set to four for the subsequent sensitivity analysis.

Table 6
Thermophysical properties of the heat pipe.

Parameters	Values
Heat pipe length (mm)	300
Effective length (mm)	120
Heat pipe diameter (mm)	7
Wick type	Sintered
Condenser length (mm)	60
Evaporator length (mm)	180
Adiabatic section (mm)	60
Operating temperature (K)	303–393
Heat transfer coefficient (W/m^{2}K)	1600

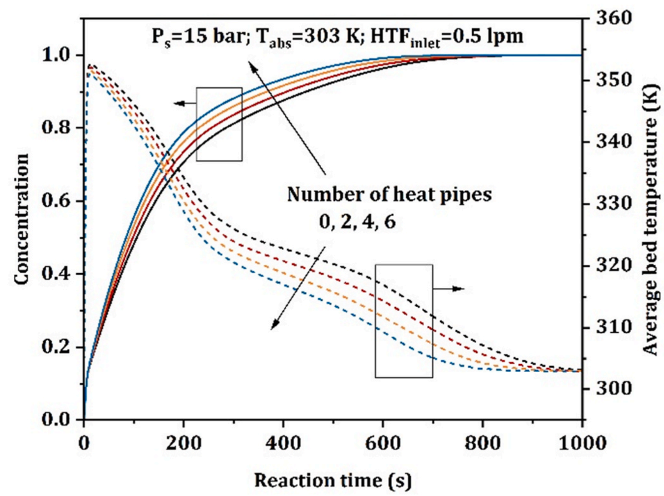


Fig. 7. Effect of number of heat pipes on hydrogen concentration and average bed temperature.

5.2.2. Effect of supply pressure

The effect of varying the supply pressure from 5 to 15 bar on the hydrogen reaction rate and average bed temperature of MHB is depicted in Fig. 8. As the driving factor for the reaction kinetics is the difference between supply pressure and the equilibrium pressure of the MHB at a given temperature [32], increasing the supply pressure will expedite the

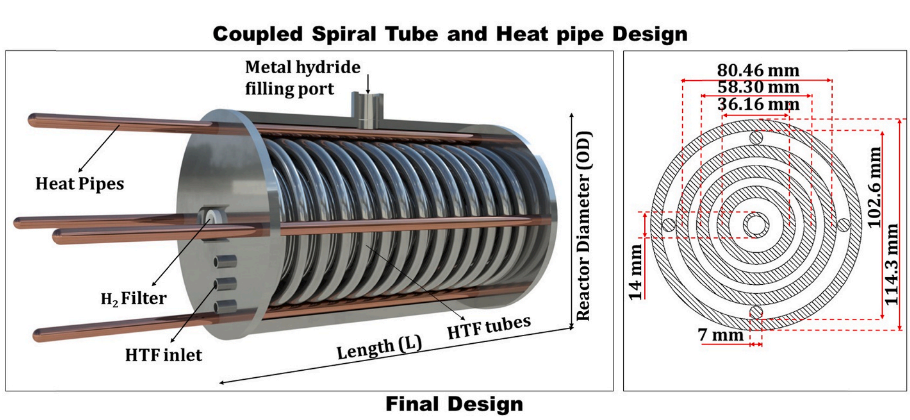


Fig. 6. Pictorial and cross-sectional view of coupled spiral tube and heat pipe design.

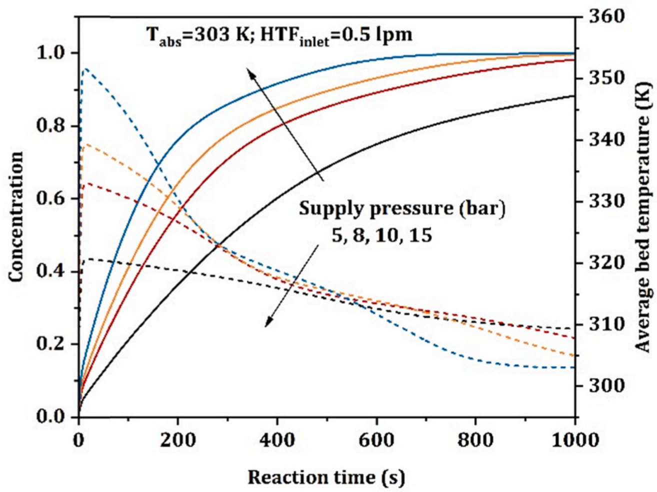


Fig. 8. Effect of hydrogen supply pressure on hydrogen concentration and average bed temperature.

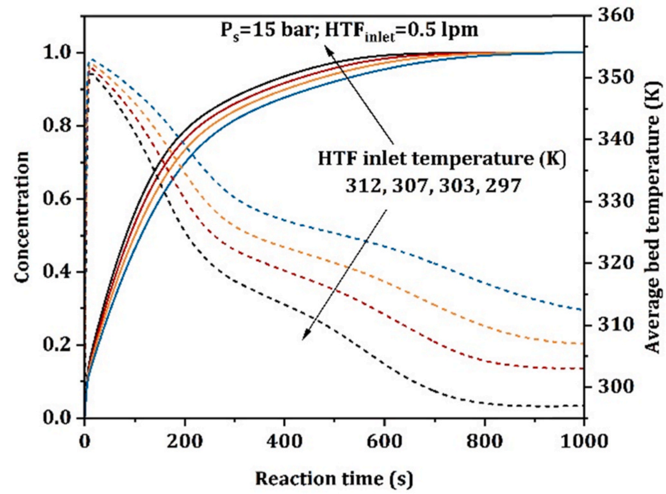


Fig. 9. Effect of cooling fluid inlet temperature on hydrogen concentration and average bed temperature.

reaction rate, which is evident from Fig. 8. For the supply pressure of 5 and 8 bar, the hydrogen storage capacity of the reactor was less than the maximum achievable weight percentage of 1.4% within 1000 s. For a supply pressure of 15 bar, the MHB can achieve 90% and 100% saturation at a period of 366 and 640 s, respectively. The peak temperature of MHB rises with increasing supply pressure due to a sudden spike in reaction rate caused by a relatively high driving force, which will decrease depending on the efficiency of the heat transfer arrangement within the system. For a supply pressure of 5 bar, the peak temperature of metal hydride is estimated to be approximately 320 K, and the hydrogen concentration is around 7% during the interval. For this condition, the temperature of the MHB hasn't reached the initial stage in the time frame of 1000 s. The maximum temperature rises for a 15-bar supply pressure is around 351 K, corresponding to a hydrogen concentration of 12%, and it gradually reverts to the initial temperature at about 930 s. Although the peak temperature is higher at higher supply pressure, it reaches the initial temperature quicker due to a faster heat transfer rate in the early phase and a lower enthalpy generation later. Increasing the hydrogen supply pressure accelerates the hydrogen sorption rate, but the discrepancy declines due to the intrinsic reaction limit caused by the interaction among driving force (P_s-P_{eq}) and heat transfer rate.

5.2.3. Effect of HTF inlet temperature

The effect of HTF inlet temperature, which is the same as the initial MHB temperature, on hydrogen absorption kinetics is illustrated in Fig. 9. The HTF inlet temperature varied from 297 to 312 K, at a flow rate of 0.5 l/min and a hydrogen supply pressure of 15 bar, which is kept constant for this study. The variation in peak temperature rise for each case is barely detectable, which are 350, 351, 352, and 353 K, respectively, resulting in a similar rate of hydriding at the early phase. However, due to the apparent higher driving force for low inlet temperatures and better heat transfer effect, the rate of hydriding accelerates later than in higher inlet HTF temperature cases. However, the decrease in saturation time with the reduction in HTF inlet temperature diminishes because of the logarithmic relation between equilibrium pressure and temperature. The temperature increment tailored in this study aids in identifying this property of the selected hydride. The time taken for the HTF temperature of 297, 303, 307, and 312 K about 90% hydrogen saturation is 333, 366, 402, and 449 s, respectively; this implies that the system achieves maximum reacted fraction in the shortest period with a higher cooling rate at low HTF inlet temperatures. The system reached complete saturation at 704 s for an inlet temperature of 297 K and 777 s for 303 K. The cases with inlet temperatures of 307 K and 312 K could not return to their initial temperatures within the stipulated time.

5.2.4. Effect of heat transfer coefficient on the exterior surface

In other characterization studies, the exterior surface of the reactor was taken as an adiabatic condition for evaluating the optimal effect of the heat transfer system arrangement. In practice, heat transfer from the reactor's exterior surface is a critical factor in determining performance. The impact of the exterior surface's heat transfer on the hydrogen reacted fraction and average bed temperature by varying heat transfer coefficient is shown in Fig. 10. The heat transfer coefficient is varied from 0 to 2000 W/m^{2}K with an increment of 500 W/m^{2}K. It is observed from Fig. 10 that, in comparison to other parameters such as supply pressure, this has a negligible effect on the reactor's heat and hydrogen mass transfer after a certain value. Beyond 500 W/m^{2}K, there was no significant effect on average absorption time, but the average MHB temperature decreased marginally to 1000 W/m^{2}K. The slight temperature variation in the metal hydride does not affect the equilibrium pressure due to the PCT characteristics, so it emphasizes that increasing the heat transfer coefficient by external means above 500 W/m^{2}K is inconsequential. For 500 W/m^{2}K, the time for reaching 90% hydrogen saturation is 216 s, and for achieving an initial average bed temperature of 303 K is 694 s; this has a percentage decrement of 31% and 25%,

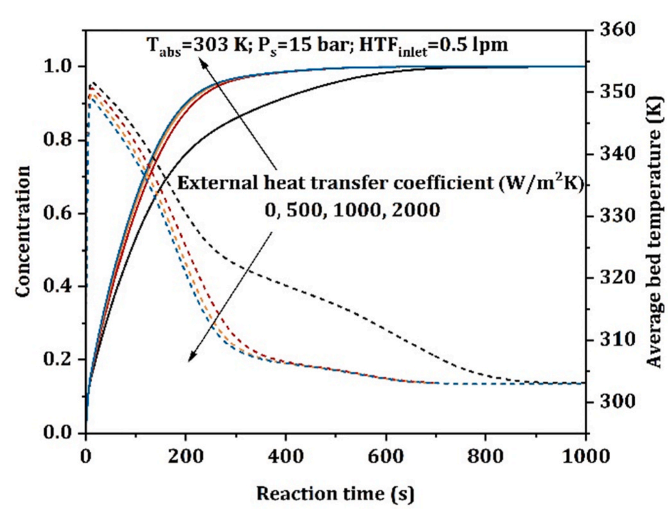


Fig. 10. Effect of heat transfer coefficient from reactor outer periphery on hydrogen concentration and average bed temperature.

respectively, concerning the reactor without any external heat transfer coefficient.

5.2.5. Effect of HTF flow rate

The flow rate of cooling fluid is another paramount parameter that affects the performance of the metal hydride reactor. But this effect is significantly less compared to the effect of supply pressure of hydrogen. The effect of the flow rate of HTF on the heat and mass rate of the alloy bed is shown in Fig. 11. Here, the flow rate is varied from 0.25 l/min to 5 l/min. This high flow rate increases fluid flow turbulence and thus the heat transfer coefficient, reducing absorption time significantly by increasing the heat dissipation rate from MHB. The changes in performance were evident up to 3 l/min; afar that value, the effect of flow rate is minimal. The variation in performance was more evident in the average bed temperature curve than in absorption time, and this is mainly due to the PCT characteristics of the hydride, i.e., the reduction in equilibrium pressure is prolonged with the variation in bed temperature. For 0.25 l/min, the time for attaining 90% hydrogen saturation is estimated as 476 s, but it is unable to revert to the minimum temperature of 303 K within 1000 s. Compared to 0.5 l/min, 3 l/min reduced 90% hydrogen saturation time and average bed temperature by 20% and 18%, respectively, and did not decrease further when the flow rate was increased above 3 l/min.

The space-time transformation of temperature and reacted fraction in the alloy domain and the temperature variation in HTF over different time scales are depicted in Fig. 12. The reaction is carried out at 15 bar supply pressure, 303 K HTF inlet temperature, and 0.5 l/min flow rate conditions and the time frames were chosen for these pictorial representations are 0, 50, 100, 250, 500, and 1000 s. The temperature of the metal hydride rises sharply at the beginning of the absorption process due to a higher reaction rate, and after that timespan, further temperature rise ceases due to the reduced reaction rate and the continued heat removal rate by the HTF. Similarly, the temperature of the HTF rises with time in the axial direction and the reacted fraction is more in the vicinity of the inlet region of HTF than the region where HTF leaves. Hence, the temperature drops and the hydrogen absorption rate decreases in the axial direction, and complete equilibrium between MHB and HTF occurs within 1000 s.

Although the exergy of the recovered heat by the heat transfer frames of the particular hydride is low, the tecno-economic comparison between the modified design and its counterpart delivers the supremacy of the reactor in terms of thermal energy retrieved. The complete economic analysis of a metal hydride reactor is dependent on sorption processes and reactor components. Therefore, an economic report based solely on

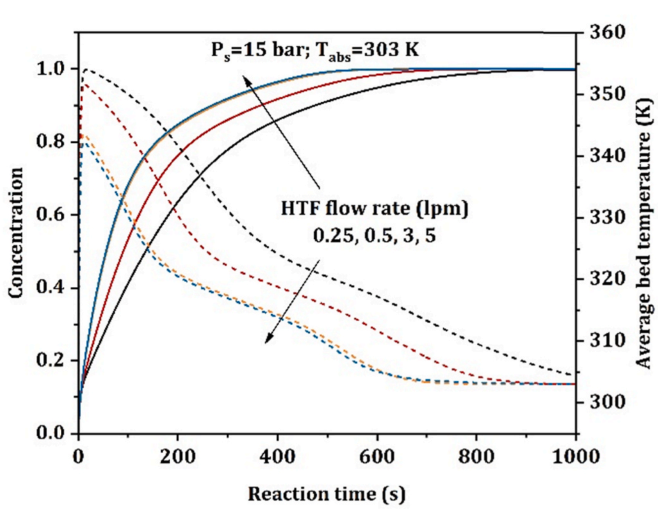


Fig. 11. Effect of heat transfer flow rate on hydrogen concentration and average bed temperature.

thermal energy savings between the spiral and modified reactor is projected to enumerate the economic features incurred by the advanced design. The economic report aims to analyze the reactor operating about 12 h/day, which is given in Table 7. The energy savings of the reactors are calculated using the thermal energy retrieved from the reactor via the heat exchanger (E_{out}/t_a). It is possible to continue the analysis for hydrogen desorption to estimate the effective energy utilization of the reactor. The economic balance equation for loss recovery is

$$\dot{C}_R + n_d \sum \dot{C}_{p,input} = n_d\left(\dot{C}_{P,output} + \dot{C}_{fuel}\right) \tag{17}$$

where $\dot{C}_R$ is the total cost of the reactor, including material, alloy, and manufacturing costs, $\dot{C}_P$ is the cost rates of power/day, $\dot{C}_{fuel}$ is the energy received from fuel distribution/day, and n_d represents the number of days at which the loss was recovered. Fig. 13 depicts the variation in net savings between the two reactors. The modified reactor can recover the difference in initial costs in 133 days and turn a profit in 1242 days. This tecno-economic comparison outlines the time and energy savings of the advanced hydride reactor design developed from the spiral heat exchanger assembly chosen based on SCR and SOER.

6. Conclusion

The comparison of different reactor designs shows that the spiral design with three tubes has a 9.5% and 12% improvement in specific charging rate and specific output energy rate, respectively, over the shell-and-tube design for the same alloy weight ratio of 0.55. In contrast, the practicality of tubular design is thwarted due to its lower alloy weight ratio. Based on the descriptive analysis, a novel reactor is developed comprising spiral tubes and heat pipes through the design progression of spiral configuration. The main conclusions outlined from the study of the modified design are as follows.

- Retrofitting heat pipes between the outer frame and outermost spiral tube will increase the reaction rate since the hydrogenation volume is more in the vicinity of the outer tube than in internal tubes.
- Increasing the number of heat pipes reduces the reaction time gradually as it acts as a heat transfer enhancer. On the other hand, adding heat pipes had a lower impact on reactors with external heat transfer arrangements (cooling jackets, fins, phase change materials etc.).
- For a supply pressure of 15 bar, the developed design with four heat pipes completed 90% hydrogen saturation of 5 kg $LaNi_5$ in 366 s when HTF was fed at 0.5 l/min at a temperature of 303 K with an adiabatic condition on the external periphery of the reactor.
- The specific charging rate and the specific output energy rate of the developed design are 0.0231 stL/min.g and 527 W/kg, respectively, which is 21.5% and 23.7% more than the reactor without heat pipes.
- The economic analysis implies that the novel reactor surpasses the net savings cost of the optimized spiral reactor in 133 days, owing to the 62 s difference in absorption and making it profitable in around 1242 days.

Furthermore, sensitivity analysis of the novel reactor indicated that it did not reach its 1.4 wt% within the entailed timeframe at lower supply pressure. Similarly, for HTF inlet temperature, increasing the temperature reduces the reaction rate, but the variation was low compared to supply pressure since the material's temperature dependency is small (i.e., negligible change in driving force). The change in average absorption concentration and bed temperature after 500 W/m^2K and 3 l/min was marginal for the proposed design. Overall, when working space and energy output are traded off equally, it can be deduced from the extensive quantitative modelling studies that the proposed novel design is an excellent choice for precise stationery and onboard thermal applications.

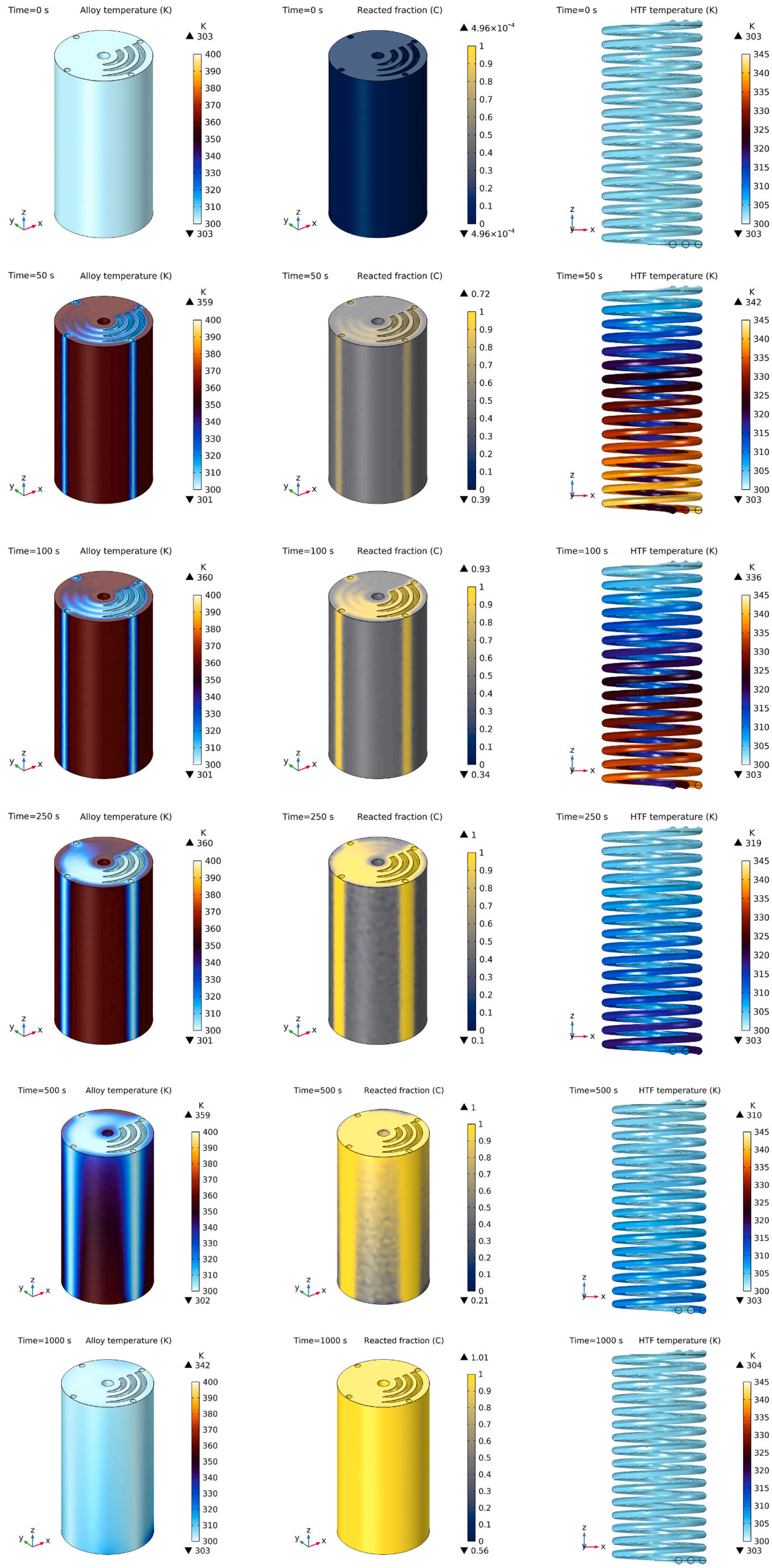


Fig. 12. Contour of alloy bed temperature, reacted fraction and HTF temperature at different time intervals (0, 50, 100, 250, 500, 1000 s).

Table 7
Economic performance of the MHHS system's absorption cycle.

Model	Material cost (₹)	Alloy cost (₹)	Manufacturing cost (₹)	Energy consumption of constant temp bath (₹/day)	Energy savings (₹/day)
Spiral	₹994.5 ($12.60)	₹19581.4 ($ 248.16)	₹30000 ($ 380.19)	₹150 ($ 1.90)	₹157.932 ($ 2.0)
Modified design	₹5794.5 ($ 73.43)	₹19581.4 ($ 248.16)	₹30000 ($ 380.19)	₹150 ($ 1.90)	₹194.463 ($ 2.46)

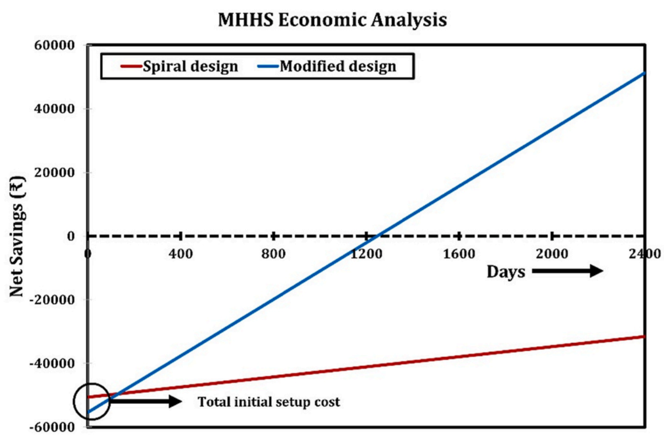


Fig. 13. Economic comparison of the selected and modified spiral designs comprising setup costs, operational costs, and payback period.

CRediT authorship contribution statement

R. Sreeraj: Conceptualization, Methodology, Study design, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft. **A.K. Aadhithiyan:** Investigation, Writing – review & editing. **S. Anbarasu:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

Symbols

T: Temperature (°C)
P: Pressure (bar)
u: Velocity (m/s)
S: H_2 absorption rate per unit volume (Kg/m^3s)
ΔS: Entropy of formation (J/mol H_2 K)
ΔH: Enthalpy of formation (J/mol H_2)
E_a: Activation energy (J/mol H_2
M: Molecular weight (kg/mol)
C_p: Specific heat capacity (J/kg K)
R_u: Gas constant (J/mol K)
t: Time (s)
q_H: H_2 absorbed from 90% saturation (stL)
E_{out}: Heat removed from 90% saturation (J)
W: Mass (kg)
t_a: Time required for 90% saturation (s)
wt%: Maximum H_2 storage capacity (%)
C_a: Reaction constant (1/s)
κ: Permeability (m^2)
d: Diameter (m)
r: Radius (m)
i: Curvature ratio
L: Length of the reactor (m)
A: Cross-sectional area (m)
h: Heat transfer coefficient (W/m^2 K)
n: number
$\dot{C}$: Cost

Abbreviations

MHB: Metal hydride bed
MHHS: Metal hydride-based hydrogen storage
HTF: Heat transfer fluid
ID: Inner diameter
OD: Outer diameter
SCR: Specific charging rate
SOER: Specific output energy rate

Greek letters

φ: Slope factor
φo: Slope Constant
ρ: Density (kg/m^3)
k: Thermal conductivity (W/m K)
β: Hysteresis factor
ε: Porosity

Subscripts

MH: Metal hydride
g: Hydrogen Gas
eq: Equilibrium
ref: reference
eff: effective
f: Heat transfer fluid
p: particle
w: Reactor wall
CT: Coolant tube
a: Alloy
i: Initial state
sat: Saturation state
s: Supply
hp: Heat pipe
c: Reactor outer surface
d: days
P: power